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AQA · LEVEL 1 · MATHEMATICS

AQA Functional Skills Mathematics Level 1

June 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4367-QP-JUN17

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 1

Data Book (Examination)

Insert

Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working you wish to have marked is written in the spaces provided in the question/answer book.

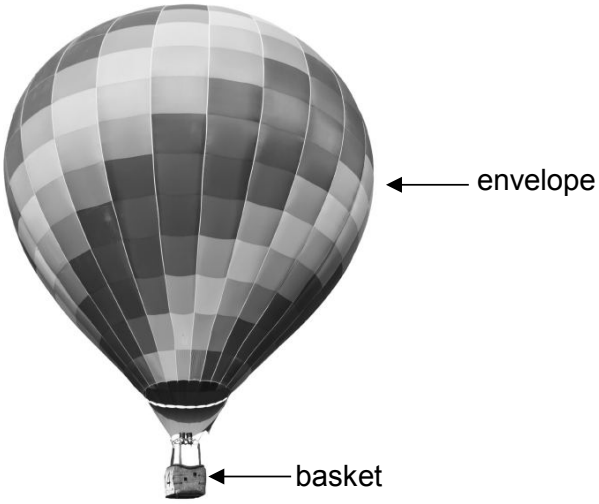
4367

QAN 500/8703/4

IB/M/Jun17/E5

Data sheet for Hot Air Balloon

A hot air balloon has an envelope and a basket.



The basket holds the pilot and passengers.

Pilot qualification

Private balloon pilots can qualify to carry passengers.
To qualify, they must

- complete 10 flights
- complete 35 hours of flying
- be the pilot of a balloon that safely rises to 5000 feet
- pass written and practical tests.

Envelope volume

The volume of the envelope is measured in m^3
An envelope with a volume of 2800 m^3 can lift a basket holding up to 5 people.

Height

Use these steps to convert height in metres to height in feet.

Step 1 Work out height in metres ÷ 3

Step 2 Work out answer to **Step 1** × 10

Data sheet for Room Makeover

Heat output of a radiator

The heat output of a radiator is measured in British Thermal Units per hour (BTU per hour).
Different types of room need radiators with different heat outputs.

The table shows the room factor for different types of room.

Room type	Room factor
Living, dining, bathroom	177
Bedroom	141
Kitchen, stairs, hallway	106

A living room has a higher room factor than a kitchen because it needs more heat.

Use these steps to work out the **minimum** heat output needed for a room.
All measurements are in metres.

Step 1 Work out length of room × width of room

Step 2 Work out answer to **Step 1** × height of room

Step 3 Work out answer to **Step 2** × room factor

The answer to **Step 3** is the minimum BTU per hour needed.

Radiator size

This table shows some radiator sizes and their heat output in BTU per hour.

Length (mm)	Heat output (BTU per hour)
500	1768
600	2122
700	2475
800	2829
900	3182
1000	3536

Example

Jenny wants a radiator for her kitchen.

For a kitchen the room factor is 106

The room has length 4 m, width 3 m and height 2.5 m

Step 1 4 × 3 = 12

Step 2 12 × 2.5 = 30

Step 3 30 × 106 = 3180

The kitchen needs a minimum of 3180 BTU per hour.

The smallest radiator Jenny should choose has length 900 mm

Buying and laying carpet

Some shops sell carpet pieces left over from a large roll of carpet.

Sometimes you can buy a piece that is exactly the right size for your room.



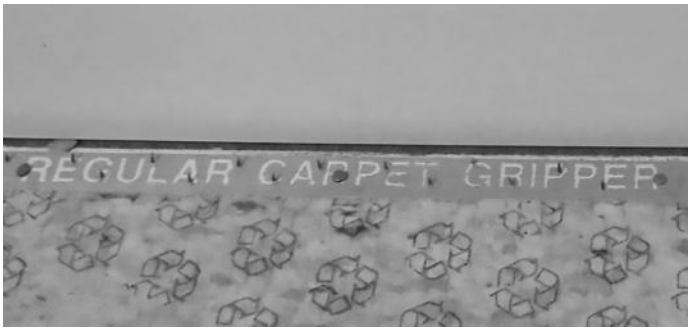
Underlay should be put on the floor first.

It is usually sold by the square metre.

It can be cut to fit the floor of the room.

The carpet is put on top of the underlay.

It is held in place using **gripper rods** around the edges of the room.



END OF DATA

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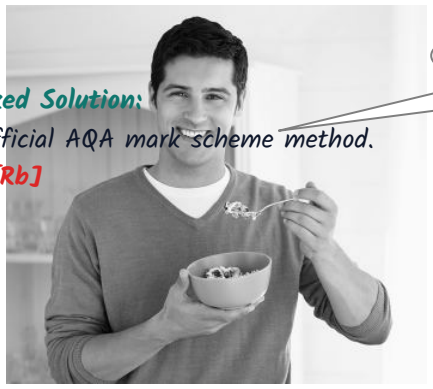
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Answer **all** questions in the spaces provided.

1 Muesli



Q1 Worked Solution:
See official AQA mark scheme method.
[B1] [Rb]

Rory

Rory finds the number of calories and the amount of sugar in different brands of muesli.

	For each serving	
Muesli	**Calories**	**Sugar**
Brand W	222	2 teaspoons
Brand X	170	$2\frac{1}{2}$ teaspoons
Brand Y	219	4 teaspoons
Brand Z	188	$2\frac{1}{2}$ teaspoons

1 (a) For a healthy diet, Rory is allowed 6 teaspoons of sugar each day.
On Monday morning, he has one serving of Brand Z.

How many **more** teaspoons of sugar is he allowed during the day?
Circle your answer.

[1 mark]

2 $3\frac{1}{2}$ $4\frac{1}{2}$ 6

Do not write outside the box

1 (b) Rory has one serving of muesli each day.

This week he has Brand Z on all 7 days.
Next week he will have Brand X on all 7 days.

He says,

"Next week, the muesli I eat will have 150 calories **less** than this week."

Is he correct?

You **must** show your working.

[4 marks]

Question 1 continues on the next page

Do not write outside the box

1 (c) For a healthy diet, Rory is allowed 2500 calories each day.

There are 260 calories in his breakfast each day.

He says,

"I have **more than** 10% of my allowed daily calories in my breakfast."

Is he correct?

You **must** show your working.

[3 marks]

Do not write outside the box

1 (d) Lara makes her own muesli.

To make 500 g of muesli she needs

350 g oats	25 g dried apricots
25 g raisins	25 g dried cranberries
40 g nuts	35 g sunflower seeds

She makes these notes.

> 100 g of oats cost 8p
>
> 100 g of dried apricots cost 72p
>
> For 500 g of muesli, the other ingredients cost a total of 95p
>
> A box of similar muesli from a shop costs £2.65 for 500 g

She says,

"For 500 g, my muesli is cheaper than the shop muesli by **more than** £1.20"

Is she correct?

You **must** show your working.

[6 marks]

14

Do not write outside the box

2 **Hot Air Balloon**

There is a **data sheet** for Hot Air Balloon.



Jo

Jo is a private balloon pilot.

She wants to qualify to carry passengers.

2 (a) Jo has already completed eight flights.

These flights lasted a total of 29 hours.

She completes two more flights.

	Day	Start time	Finish time
Flight 1	Friday	1030	1300
Flight 2	Saturday	1600	2000

Q2 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Jo says,

"I have now completed the hours I need to qualify."

Is Jo correct?

You **must** show your working.

[4 marks]

Do not write outside the box

2 (b) On one of the flights, the balloon reaches a height of 1620 **metres**.

Show that this height is more than 5000 **feet**.

Use the steps on the data sheet.

[3 marks]

†2 (c) Jo rents a balloon with an envelope volume of 3200 m^3
A balloon can lift 1 kilogram for every 4 m^3 of balloon volume.

Work out the maximum number of kilograms this balloon can lift.

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

Do not write outside the box

2 (d) Jo rents the balloon for a year.

It can carry the pilot and four passengers.

She makes these notes about the flights and costs for the year.

> The flights will start on 1 April and end on 31 October.
>
> There will be 8 flights each month.
>
> I will charge each passenger £145 for a ticket.
>
> My costs for the year will be
>
> £5700 to rent the balloon
>
> £2250 for fuel.

Jo says,

"If I sell 4 tickets for every flight, I will make a profit of **more than** £25 000"

Is she correct?

You **must** show your working.

[7 marks]

17

Do not write outside the box

3 Canteen Survey

Betty is the manager of the college canteen.



I want to improve the service we offer.
To help me I will carry out a survey.

Betty

20 students completed Betty's survey form.
Here is one of the completed forms.

Canteen Survey

A Circle the number of days that you used the canteen last week.

0 1 (2) 3 4 5

B Tick a box to show which of these sandwiches you like best.

Egg and Cress	
Bacon, Lettuce and Tomato	✓
Chicken Salad	

C The canteen is thinking of selling sushi.
Tick a box to show what type of sushi you would like best.

Salmon	
Vegetable	✓
Tuna	
None	

Do not write outside the box

3 (a) The mean number of days the 20 students used the canteen that week was 2.5

Betty put some more seats in the canteen.

The following week she asked the same 20 students question **A** again.

Here are the answers.

Q3 Worked Solution:

Marks are awarded for demonstrating the following interrelated process skills.

Representing

0 3 5 3 2 2 0 3 3 5

Selecting the mathematics and information to model a situation.

3 4 4 5 3 1 2 2 4 4

R.1

Betty says,

"The mean number of days the students used the canteen has increased."

Is she correct?

You **must** show your working.

[4 marks]

Candidates recognise that a situation has aspects that can be represented using mathematics.

R.2

Candidates make an initial model of a situation using suitable forms of representation.

R.3

[Method & Accuracy]

Do not write outside the box

The table shows information about the answers to question **B**.
It also shows the price of each type of sandwich.

Sandwich	**Number of people**	**Price**
Egg and Cress	5	£2
Bacon, Lettuce and Tomato	10	£3
Chicken Salad	5	£3
	Total = 20	

3 (b) Betty makes 80 sandwiches each day.

She uses the survey results to work out how many of each sandwich to make.

Betty works out how much money she could take from selling the sandwiches.

She says,

"If I sell all 80 sandwiches I will take **more than** £200 each day."

Is she correct?

You **must** show your working.

[6 marks]

Do not write outside the box

3 (c) The table shows information about the answers to question **C**.

Type of sushi	**Number of people**
Salmon	3
Vegetable	7
Tuna	4
None	6
	Total = 20

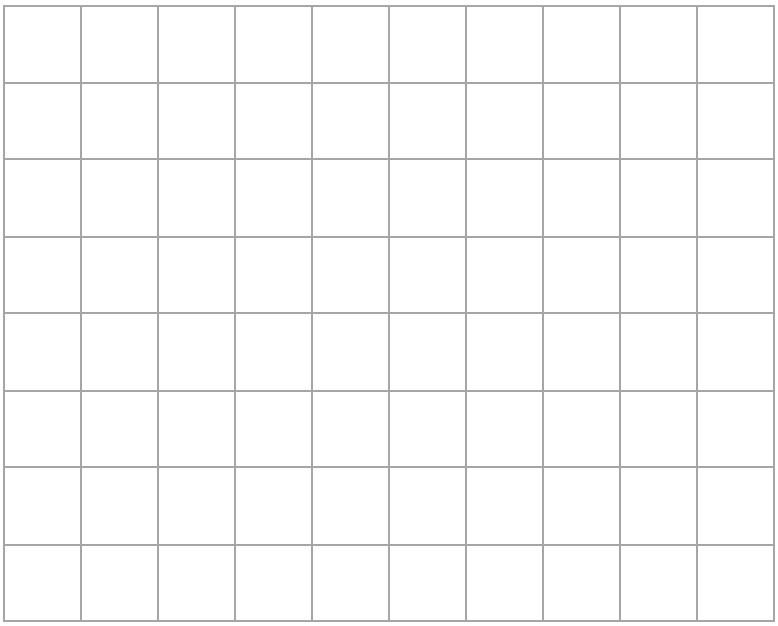
Betty wants to show this information in a diagram.

Draw a pictogram or a bar chart that Betty could use.

Use the space below **or** the grid opposite.

[3 marks]

Do not write outside the box



Turn over for the next question

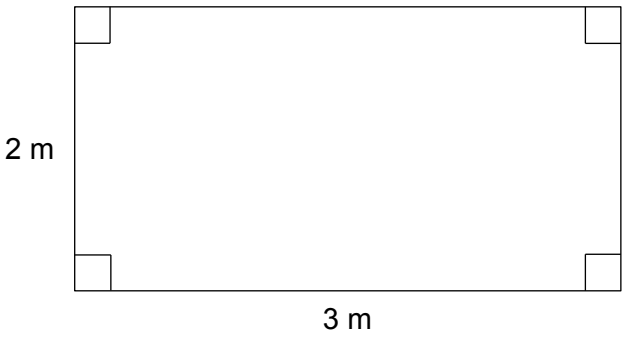
13

Do not write outside the box

4 **Room Makeover**

There is a **data sheet** for Room Makeover.

Here is a sketch of Tom's bedroom floor.



Not drawn accurately

4 (a) The gripper rods in one pack have a total length of 10.6 metres.

Show that one pack has enough gripper rods to go around the edge of the bedroom.

[2 marks]

Q4 Worked Solution:

MI for one pair

multiplied

their 56 × 4 or 224

total number of

tickets

4 (b) The bedroom has height 2.5 metres.

Tom wants a new radiator.

Work out the smallest length of radiator he should buy.

You **must** show your working.

[5 marks]

25 000 + 5700 + 2250 or 32 950

Income needed to make at least 25000

profit

their 32 950 ÷ their 224

their 32 950 ÷ 145

[MI] [Aa] [MI] [Aa] [MI] [Aa]

Do not write outside the box

4 (c) Tom is buying a carpet for his bedroom.

He buys this carpet piece, which fits the room exactly.



Carpet piece
Only £95

The shop sells

underlay for £3.99 per square metre
gripper rods for £5.79 per pack.

Tom buys

the carpet piece
enough underlay for the bedroom
one pack of gripper rods.

He says,

"I can buy all these items for **less than** £125"

Is he correct?

You **must** show your working.

[5 marks]

Turn over for the next question

Do not write outside the box

†4 (d) Tom hires Ali to paint the bedroom.

Ali charges £28 per hour, plus £55 for paint.

He works for 6 hours.

How much does Ali charge in total?

[3 marks]

Check your answer.

Show how you have done your check.

[1 mark]

16

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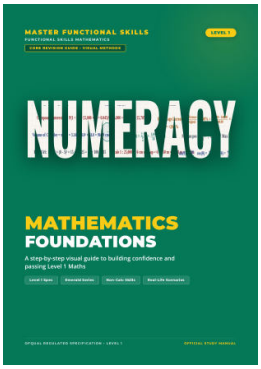
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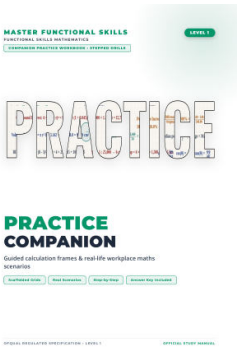


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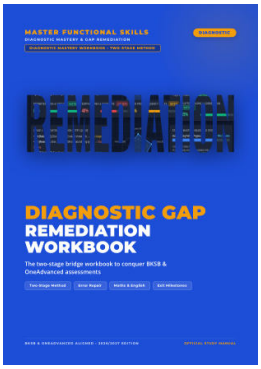


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